Yassa Ashraf

Software Engineer

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SUMMARY

Software Engineer specializing in full-stack and cloud-native systems with production experience across React, NestJS, Node.js, ASP.NET, and AWS/OCI. Delivers scalable, observable services and data-driven features with a focus on clean architecture, performance, and DX. Proven impact in startup and enterprise environments, leading cross-functional initiatives and shipping customer-visible outcomes.

EDUCATION

Bachelor of Computer Science, Major in Software Engineering — German International University, Cairo Oct 2021 – Jul 2025

BACHELOR THESIS

Using Explainable AI Algorithms for Material Recognition — Grade: A+

- Built ChemBERTa-based material recognition with post-hoc explanations; enabled traceable predictions for domain experts.
- Deployed via Flask APIs on AWS (SageMaker, EC2, S3) with autoscaling and CI/CD; optimized inference p95 latency.
- Delivered a Next.js dashboard for real-time inspection of attention maps and feature attributions.

CERTIFICATIONS

Oracle Cloud Infrastructure 2025 Certified Foundations Associate — Issued Mar 2025 Verify Credential

SKILLS

- **Languages:** TypeScript, JavaScript, Python, Java, SQL, C++
- **Frontend:** React, Next.js, Angular, React Native, Tailwind
- **Backend:** Node.js, NestJS, ASP.NET Web API, Express, GraphQL, REST
- **Data:** PostgreSQL, MongoDB, Redis, Prisma, Drizzle ORM
- **Cloud/DevOps:** AWS (Lambda, ECS/EKS, S3, EC2, SageMaker), OCI, Docker, Kubernetes, CI/CD
- **Practices:** Clean Architecture, Microservices, DDD, JWT/OAuth2, Testing (Jest)

TOOLS

- GitHub Actions, Serverless Framework, Terraform (basics), pnpm/npm, ESLint, Prettier
- Postman/Hoppscotch, pgAdmin, Prisma Studio, Grafana basics, CloudWatch
- Linux, Bash, Nginx, Vite, Webpack

SOFT SKILLS

Problem Solving, Leadership, Collaboration, Communication, Adaptability, Time Management, Creativity, Attention to Detail

EXPERIENCE

AI Software Engineer — VIXI AI (Remote) United Arab Emirates | Sep 2025 – Present

- Designed and shipped NLP pipelines for enterprise clients, integrating Transformers with domain-specific prompts and retrieval.
- Implemented containerized inference services with autoscaling and observability, improving stability and cost efficiency.
- Partnered with product and data teams to prioritize features; reduced iteration cycles by introducing preview environments.

Software Engineer — Xylem Software (Remote) Cairo, Egypt | Jun 2025 – Present

- Delivered web and mobile features in a monorepo (React, React Native, Node.js, Express) with shared components and CI.
- Led development for an analytics platform (Snap Analytics) including ingestion APIs and interactive

dashboards.
- Elevated performance via query optimization and API caching, improving key views' load time.

Associate — Vodafone Intelligent Solutions (VOIS) Cairo, Egypt | Jul 2025 – Present

- Enhanced ASP.NET Web API and Angular modules in an N-tier architecture; introduced advanced filtering and pagination.
- Refactored reusable components and improved error handling and logging; increased reliability and maintainability.
- Supported stakeholder demos and UAT, aligning deliverables with business requirements and security standards.

Intern — Vodafone Intelligent Solutions (VOIS) Cairo, Egypt | Feb 2025 – Jul 2025

- Contributed to Web API enhancements and Angular UI refinements; improved p95 response times for critical endpoints.
- Implemented unit tests and participated in code reviews, raising baseline code quality.

Mid-Level Backend Engineer — Trevi (Hybrid) Cairo, Egypt | Oct 2024 – Aug 2025

- Architected wallet/loyalty microservices with NestJS and Prisma on AWS (EC2, Lambda, S3), serving high transaction volumes.
- Integrated cashback and promotion engines; delivered measurable retention uplift and streamlined campaign operations.
- Maintained high availability with robust health checks, backoff/retry, and metrics for SLA tracking.

Software Engineer — MM Solutions (Remote) Loveland, CO, USA | Jul 2024 – Mar 2025

- Led web-integration initiatives; improved UI load time and optimized bundle size and caching headers.
- Streamlined dispatch workflows with Amazon Freight partners; reduced cycle times through automation.

Full Stack Developer — Kinetics Egypt (On-site) Cairo, Egypt | Aug 2024 – Oct 2024

- Delivered AWS-based solutions (EC2, S3) with IaC fundamentals and hardened configurations.
- Built features in coordination with product, meeting milestones ahead of schedule.

Full Stack Developer (Freelance) — Fiverr Remote | May 2023 – Oct 2023

- Built custom web apps, stock-prediction dashboards (Python + Robinhood API), and 2D games (Java/C++) with 5-star ratings.
- Managed end-to-end delivery: requirements, implementation, testing, and handover.

Full Stack Developer — Instatus Cairo, Egypt | Jul 2022 – Sep 2022

- Developed and optimized TypeScript/Next.js applications, improving reliability and performance.

SELECTED PROJECTS

ECS-Help Desk — ecs-project-omega.vercel.app MERN, Tailwind, Team Lead

- Ticketing with roles, SLAs, and status transitions; real-time updates and analytics.
- Improved response times and agent productivity through streamlined flows and targeted metrics.

Cairo Metro Reservation System Node.js, Express, Bootstrap

- Built responsive frontend and booking APIs with validation and secure session handling.
- Implemented schedule search, fare rules, and receipt generation.

E-commerce for Plastic Pallets NestJS, Next.js, Kafka, Microservices

- Designed catalog, order, and payment services with Kafka for async communication and resilience.
- Implemented admin dashboards and inventory management with role-based access controls.

Bookstore Application AWS Lambda, API Gateway, DynamoDB, S3

- Serverless CRUD and search; S3 static hosting; CI for infra and app updates.
- Observability with CloudWatch logs and alarms for error spikes.

Task Tian GraphQL, Prisma, Next.js

- Task prioritization, labels, and filters; optimistic UI updates and pagination.
- Role-based permissions and granular resolvers with schema-first design.

Bachelor Portal Flutter, SQL Procedures

- Implemented student workflows with transactional guarantees; optimized queries and indexing.

Operating System Project Java

- Implemented thread management, synchronization primitives, and scheduling strategies for concurrency scenarios.

ADDITIONAL PROJECTS

Beity App — Cloud Kitchens Delivery Management System NestJS, Clean Architecture, AWS, PostgreSQL, React Native

- Led technical design and implementation of a multi-role platform for sellers and customers.
- Architected modular backend with NestJS, PostgreSQL, and Clean Architecture for scalability.
- Deployed on AWS with CI/CD pipelines and secure multi-tenant data management.
- Delivered React Native mobile app for order placement, tracking, and seller management.

Big Data ETL Pipeline PySpark, AWS Glue, PostgreSQL, S3

- Designed and implemented distributed ETL workflows for large-scale hotel pricing datasets.
- Processed millions of records with PySpark transformations, joins, and aggregations.
- Automated ingestion from PostgreSQL into AWS S3 and Glue catalog for downstream analytics.
- Optimized Spark jobs with partitioning, predicate pushdown, and caching to reduce runtime.

E-commerce Mobile App Flutter, Firebase, Firestore

- Built catalog, cart, and checkout with secure auth; integrated analytics and crash reporting.

Clash of Clans-style 2D Game Java Swing

- Implemented game loop, collision detection, and resource management using OOP principles.

Super Mario-style Game C++

- Developed tilemaps, physics, and sprite animation with clean separation of rendering and logic.

TECHNICAL HIGHLIGHTS

- **Architecture:** Microservices with event-driven communication, API gateways, background jobs, and idempotency.
- **Performance:** Query optimization, caching strategies, asset/code splitting, and CDN usage.
- **Security:** JWT/OAuth2, input validation/sanitization, secrets management, and least-privilege IAM policies.
- **Quality:** CI pipelines with linting, tests, type checks, and preview deployments.
- **Observability:** Structured logging, metrics, alerting, and dashboards for SLIs/SLOs.

LANGUAGES

English: Professional Proficiency — Arabic: Native Proficiency — Deutsch: Limited Working Proficiency